VS

VidyaSpark

The Intellectual Forest

AI-Powered Ignator Training & Education Platform

Presented by Hari Sridhar and Levi Belnap

CRO, Agastya International Foundation | Texas A&M Mays Business School

THE CHALLENGE

Training Excellence at Scale

Agastya International Foundation has 1,000 Ignators delivering hands-on science education to millions of children across India.

The hardest problem: How do you ensure every single Ignator delivers magic in every single classroom, every single time?

Traditional training reaches each Ignator once. VidyaSpark lets them practice endlessly.

1,000
Ignators

100M
Children Goal

24/7
Training Need

THE SOLUTION

VidyaSpark: The Virtual Tapovan

An AI-powered virtual classroom where Ignators practice teaching interactive science experiments to 5 AI students who behave like real children.

Unlimited Practice

Teach the same module again
and again. AI students reset.

Available 24/7

Practice from anywhere, anytime. No scheduling.

5 Unique Students

Each AI student challenges different teaching skills.

Instant Feedback

AI or human mentors give actionable coaching.

HOW IT WORKS

Learn. Practice. Ignite.

01

LEARN

The Ignator reviews the ABL module, sees the experiment materials, and understands the learning objectives before entering the virtual classroom.

02

PRACTICE

The Ignator teaches the experiment to 5 AI students on an interactive canvas. AI students ask questions, get confused, and challenge — just like real children.

03

IGNITE

After the session, the Ignator receives a detailed report with scores, specific feedback, and coaching. Then they do it again — better each time.

Meet the 5 AI Students

P

Priya

13, Curious

Asks genuine why questions.
Pushes the Ignator to explain deeper.

R

Ravi

14, Skeptic

Demands proof, challenges claims.
Refuses answers without evidence.

L

Lakshmi

12, Shy

Knows the answers but will
not speak unless directly invited.

A

Arjun

14, Disengaged

Bored, makes jokes, distracted.
Can be won back with the right hook.

M

Meena

13, Rote Learner

Repeats textbook answers without real understanding.

The Interactive Experiment Canvas

Each ABL module from Agastya's handbooks becomes a visual, interactive experiment on the canvas.

EXPERIMENT CANVAS

The Ignator narrates: "I'm placing the brick flat on the rubber sheet"

The canvas animates the experiment step by step.

AI students see the result and react naturally:
Priya: "Why did it go down a little?"
Ravi: "That's nothing. Try it the other way."

- Narrate and Animate — the canvas responds to the Ignator's words
- Pre-built experiment sequences for each ABL module
- AI students react based on what's visible on the canvas
- Mirrors the real classroom experience

LIVE PLATFORM

It's Already Built and Running

vidyaspark.onrender.com

8 fully functional modules live today, across 4 subjects. All sourced directly from Agastya's ABL handbooks. Real AI students, real assessments, real scoring.

LIVE DEMO

1. Select any ABL module from the library
2. Enter the Tapovan — the virtual classroom with 5 AI students
3. Teach the experiment using the interactive canvas
4. Receive a comprehensive assessment report with composite scoring

MODULE LIBRARY

8 Modules Live Today

PHYSICS

Force & Pressure

Brick on rubber sheet — exploring how area affects pressure

Sound

Vibrations, frequency, and how sound travels

CHEMISTRY

States of Matter

Solids, liquids, gases — particle arrangement

Separation Techniques

Filtration, evaporation, chromatography

MATHEMATICS

Triangles

Types, properties, and the angle sum property

Quadrilaterals

Classification and properties of quadrilaterals

BIOLOGY

Cells

The fundamental unit of life — onion peel experiment

Photosynthesis

How plants make food using sunlight

Architecture supports 100+ ABL modules across all Agastya handbooks

Rigorous Assessment & Scoring

How Assessment Works

1. Ignator completes the teaching session with all 5 AI students
2. AI generates 10 comprehension questions based on the ABL module
3. Each AI student answers based on their actual engagement during the session
4. Students who were more engaged score higher — just like real classrooms

COMPOSITE SCORE

70%
Student Achievement Score

+

30%
Ignator Confidence Rating

THE VISION

Three Phases to Transform Education

PHASE 1

NOW

THE SIMULATOR

Ignators practice teaching ABL modules to 5 AI students. They receive mentor feedback. They practice until mastery. This is the research phase.

PHASE 2

NEXT

THE CLASSROOM

The same interactive canvas becomes the teaching tool for real students. Every government school with a screen becomes a science lab.

PHASE 3

SCALE

THE PLATFORM

VidyaSpark deploys across Andhra Pradesh and beyond. 100+ interactive modules reaching thousands of schools through trained Ignators.

A World-First Research Opportunity

The first rigorous, randomized study of AI-powered teacher training in India — or anywhere.

EXPERIMENTAL DESIGN

Treatment: VidyaSpark-trained Ignators

Control: Traditionally trained Ignators

Measure: Student learning outcomes in real classrooms

Sample: 50+ Ignators over 3 months

Variables: Teaching quality, engagement, concept retention

RESEARCH IMPACT

- Publishable in top education and technology journals
- Evidence base for government adoption across Indian states
- Global implications for AI in teacher training
- Positions Agastya as a research-driven innovator

Proposed Path Forward

01 INCORPORATE
Establish VidyaSpark as a company — a technology venture with a social mission.

02 COLLABORATE
Partner with Agastya to build ALL 100+ ABL modules from the complete handbook library.

03 PILOT
Run a rigorous 3-month experiment with 50 Ignators. Measure real impact on student learning.

04 PUBLISH
Publish research results in top journals. Build the evidence base for AI-powered teacher training.

05 SCALE
Deploy across Andhra Pradesh. Present to the Chief Minister with proven results.

FUNDING

Funding the Collaboration

How could funding work for this partnership?

Agastya Co-Investment

Direct investment from Agastya Foundation in VidyaSpark development and deployment.

Government Education Grants

Andhra Pradesh state education technology funding for classroom deployment.

Research Funding

NSF grants, education foundations, and academic research partnerships.

Corporate CSR

Indian corporate social responsibility partnerships (tech companies, education firms).

Revenue from Deployment

State-level licensing and per-school deployment revenue at scale.

THE GOAL

100 Million Children

1 Million Teachers

Powered by the Intellectual Forest

VidyaSpark makes Agastya's 2032 vision achievable by giving every Ignator a tireless practice partner.

Let's Build This Together

Hari Sridhar and Levi Belnap

NEXT STEPS

Discuss collaboration structure and funding
Plan the 50-Ignator pilot experiment
Build the complete ABL module library together

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